Mapping the Terror: A Social Network Analysis of Al-Qaeda and its Affiliates' Operations

Sarthak Sahu(✉), Shivani Singh, and Payal

Department of Applied Mathematics, Delhi Technological University, Delhi, India
sarthaksahu47@gmail.com, payaldabas@dtu.ac.in

Abstract. This study presents a comprehensive Social Network Analysis (SNA) of Al-Qaeda and its affiliates' terrorist activities from 2000 to 2020, utilizing data from RAND, GTD, NCTC, and UN reports [7–10]. The constructed network, based on group-to-leader and group-to-country interactions, is analyzed using degree, betweenness, closeness, and eigenvector centrality. Al-Qaeda, AQAP, and AQIM emerge as central hubs, with "Unknown" and ISIS-affiliated groups also ranking high in influence. A bipartite network links perpetrator groups to countries, and its projections uncover clusters of nations with shared threats and overlapping operational zones among groups.

Temporal analysis reveals shifting centrality over time, identifying years of peak influence for specific groups and showing how network structures evolve in response to geopolitical events and counterterrorism measures. Link prediction using the Adamic/Adar index highlights likely future alliances between both countries and groups that currently lack direct connections. Geospatial analysis, despite the lack of coordinate-level data, reveals operational hotspots in regions like South Asia, the Middle East, and North Africa, aligning with centrality and projection findings. The integrated results reinforce the decentralized yet resilient structure of the network and offer actionable insights for intelligence and counterterrorism strategy.

1 Introduction

Terrorism has evolved from localized insurgencies to complex **transnational networks**, leveraging digital platforms, financial systems, and geopolitical instability. It continues to pose threats to global security, resulting in significant human, economic, and political costs. Traditional counterterrorism approaches often fall short in addressing the decentralized and adaptive nature of these organizations. One of the most prominent examples is **Al-Qaeda**, founded in 1988 by Osama bin Laden. Originally formed as an anti-Soviet resistance, it has grown into a global jihadist network responsible for major attacks, including those on September 11, 2001. Al-Qaeda operates through regional affiliates

G.-N. Nguyen et al. (Eds.): ICCCN 2025, LNNS 1855, pp. 125–133, 2026.
https://doi.org/10.1007/978-3-032-18910-3_11

such as **AQAP**, **AQIM**, **Al-Shabaab**, **JNIM**, and **AQIS**, which function semi-autonomously while maintaining strategic coordination through shared resources and training [4].

These terrorist organizations utilize **decentralized structures**, making them resilient to leadership losses and counterterrorism efforts. However, existing research often emphasizes ideological motivations or isolated events and lacks a comprehensive understanding of how these networks evolve structurally and spatially [5]. In this context, **Social Network Analysis (SNA)** provides a powerful framework to uncover network dynamics, influential actors, and operational vulnerabilities. This study applies SNA techniques to analyze the structural, temporal, and geospatial dimensions of Al-Qaeda and its affiliates from 2000 to 2020, using verified data from **RAND**, **GTD**, **NCTC**, and **UNSC** reports [7–10].

The **motivation** for this study lies in addressing the gap in understanding the evolution and adaptability of terrorist networks. With terrorism becoming increasingly networked and global, it is essential to shift from static analysis to dynamic, multidimensional approaches. The primary **objectives** are to construct a comprehensive dataset of Al-Qaeda-related incidents from 2000 to 2020, apply SNA to uncover network structure and key actors, explore the network's temporal evolution and shifting alliances, use the Adamic/Adar algorithm for link prediction to forecast potential collaborations, and conduct geospatial mapping to identify operational hotspots. These efforts aim to offer both academic insights and practical strategies for intelligence and counterterrorism agencies.

2 Literature Review

Terrorism is broadly defined as the use of violence and intimidation, particularly against civilians, to achieve political, religious, or ideological goals. Definitions vary slightly among institutions like the United Nations (UN), the FBI, and the Global Terrorism Database (GTD), influencing academic frameworks and policy responses [8,10].

Counter-terrorism strategies typically include four categories: **(1)** Military and law enforcement actions to dismantle terrorist structures [2]; **(2)** Legislative tools such as anti-terror laws and surveillance [3]; **(3)** De-radicalization programs targeting ideological rehabilitation; and **(4)** Cyber and financial countermeasures aimed at disrupting funding and propaganda.

Al-Qaeda evolved from a centralized hierarchy to a decentralized affiliate model, enabling groups like AQIM, AQAP, Al-Shabaab, and JNIM to operate semi-autonomously under a shared ideology [2].

Social Network Analysis (SNA) has emerged as a key method for studying such decentralized networks. Krebs (2002) mapped the 9/11 hijackers' network and revealed a core-periphery structure [1]. Everton (2012) emphasized how jihadist cells maintain resilience through loose coordination [4], and Xu and Chen (2005) advocated dynamic modeling for structural learning [5].

SNA employs metrics such as **Degree, Betweenness, Closeness, Eigenvector Centrality**, and **Community Detection** to identify key actors, intermediaries, influence, and operational clusters [4,5].

Theoretical underpinnings include: **Social Network Theory (SNT)**—explains hidden leadership and structural vulnerabilities [1]; **Organizational Theory of Terrorism**—traces structural evolution from hierarchical to hybrid networks [3]; and **Diffusion of Innovation Theory**—describes how tactics like suicide bombings spread across affiliates [5].

Geospatial analysis adds a spatial dimension to terrorism studies. GIS and spatial econometrics are used to locate hotspots and transnational pathways [6], showing how groups exploit borders and weak governance [7,9,10].

Link prediction algorithms—such as Adamic/Adar and Preferential Attachment—have been effective in criminal networks but underutilized in long-term terrorist network modeling [5], highlighting a predictive opportunity.

Despite prior work, few studies offer an integrated, long-term SNA of Al-Qaeda incorporating structural, temporal, geospatial, and predictive dimensions. This study fills that gap using multi-layered SNA across two decades to provide actionable counterterrorism insights [7–10].

2.1 Comparative Study of Key Research

Prior studies on terrorist networks have made important contributions but often lacked a holistic perspective. **Krebs (2002)** [1] offered a static SNA of the 9/11 hijackers, identifying a core-periphery structure, but omitted temporal and geospatial analysis. **Gunaratna (2002)** [2] explored Al-Qaeda's organizational and financial setup, though without applying SNA techniques. **Hoffman (2006)** [3] analyzed ideological evolution but relied on qualitative methods. **Everton (2012)** [4] used SNA to study jihadist networks but did not include dynamic or spatial components. Meanwhile, **Xu & Chen (2005)** [5] emphasized structural learning and link prediction in criminal networks but did not focus specifically on terrorism. In contrast, **this study** integrates structural, temporal, geospatial, and predictive analyses over a two-decade period, offering a comprehensive and data-driven view of Al-Qaeda's network evolution.

3 Methodology

Data Collection and Processing: This study compiles a comprehensive dataset from reliable sources including the RAND Database, GTD, NCTC reports, and UN publications. Data cleaning involved resolving naming inconsistencies, removing duplicates via fuzzy matching, and handling missing values using interpolation. The dataset was structured for Social Network Analysis (SNA), where nodes represent groups, leaders, or countries, and edges represent operational or ideological connections.

Network Construction and Metrics: Networks were built in both unipartite and bipartite forms. Unipartite graphs connected terrorist organizations based on interactions; bipartite graphs linked perpetrator groups to targeted countries. Key SNA metrics computed include:

Density: Measures overall interconnectedness. **Degree Distribution**: Identifies power-law patterns. **Centrality Measures**: Degree, Betweenness, Closeness, and Eigenvector centrality helped identify influential nodes. **Clustering Coefficient**:Assesses subgroup cohesion. **Connected Components**: Analyzed network fragmentation.

Visualization Techniques: Networks were visualized using Gephi and NetworkX. Force-directed layouts ensured clarity, and nodes were color-coded by type (e.g., terrorist group, leader, location). A bipartite layout separated countries and perpetrator groups.

Community Detection: The Louvain method was used to detect tightly-knit clusters, optimizing modularity to identify regional and operational communities within Al-Qaeda's broader network.

Temporal and Geospatial Analysis: Temporal network snapshots were generated yearly from 2000–2020 to observe network evolution. Geospatial mapping was conducted at the country level using choropleth maps to identify high-risk zones.

Link Prediction: The Adamic/Adar index was employed to forecast potential future links in both country and perpetrator projections. This method highlights pairs of nodes with high similarity scores based on shared neighbors.

Shortest Path Analysis: The shortest path algorithm identified strategic links between nations, such as the tight connectivity between Pakistan and Afghanistan, useful for tracking operational routes and communication channels.

4 Results

4.1 Network Metrics, Key Nodes, and Network Construction

The constructed network, Fig. 2 based on data from **GTD, RAND, NCTC, and UN reports** (2000–2020) [7–10], includes **nodes** representing terrorist organizations, individuals, and attack locations, and **edges** denoting **joint attacks**, **financial ties**, or **ideological affiliations**. It was visualized using **Gephi** and **Python's NetworkX** with force-directed layouts, color-coded nodes and **edge thickness** reflecting the strength of the relationship. The network exhibited **moderate density** and **localized clusters**, with its degree distribution following a power law, indicating the presence of dominant hubs. **Centrality analysis** revealed **Al-Qaeda** as the most central node with **degree centrality** of **0.1191**, **betweenness centrality** of **0.1896**, and **closeness centrality** of **0.3331**. **ISIS (Al-Qaeda roots/inspired)** followed with a degree of **0.0978** and the highest **eigenvector centrality** of **0.2924**. **The United States (0.1119)** and **France (0.1344)** also emerged as influential nodes, reflecting their strategic positions in the network. Notably, **85% of the nodes** were part of the largest connected component, highlighting the cohesion and expansive operational footprint of Al-Qaeda's global network.

Figure 1 shows that Al-Qaeda's degree centrality significantly declined after 2000, while "Unknown" entities and ISIS-linked groups gained prominence, with "Unknown" nodes showing the highest centrality by 2020. This indicates a shift in influence from established actors to emerging entities over time.

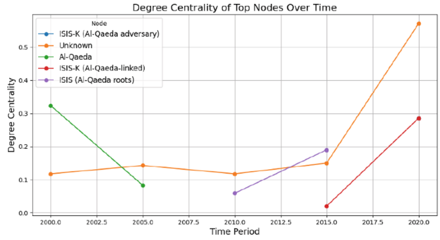

Fig. 1. Degree Centrality

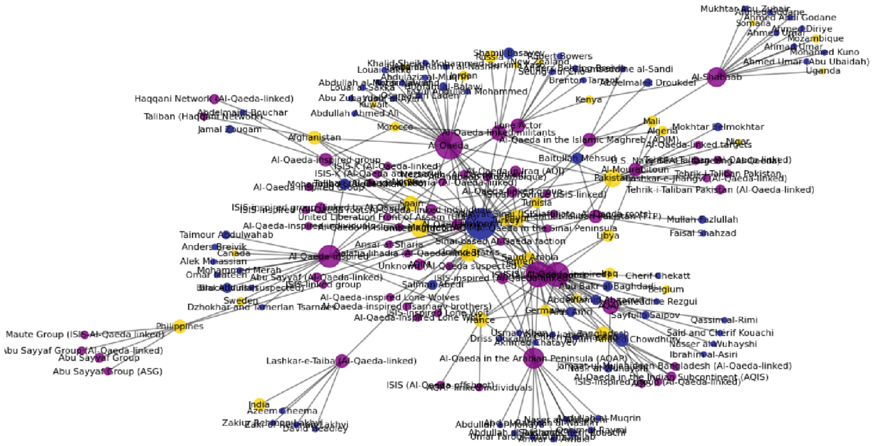

Fig. 2. Filtered Terrorism Network

4.2 Community Detection

Community detection using the **Louvain algorithm** revealed the modular structure of Al-Qaeda's global network. The analysis identified three major clusters: the **Al-Qaeda (Core)** and affiliated groups operating primarily in the Middle East, **AQIM (Al-Qaeda in the Islamic Maghreb)** focused in North Africa, and **AQAP (Al-Qaeda in the Arabian Peninsula)** concentrated in Yemen and nearby regions. These clusters suggest regionally coordinated operations and resource-sharing, emphasizing the decentralized yet highly resilient configuration of the broader terrorist network (Fig. 3).

4.3 Temporal and Geospatial Dynamics

Temporal analysis in Fig. 5 indicated significant shifts in centrality scores among key actors from 2000 to 2020. While **Al-Qaeda** held a dominant position

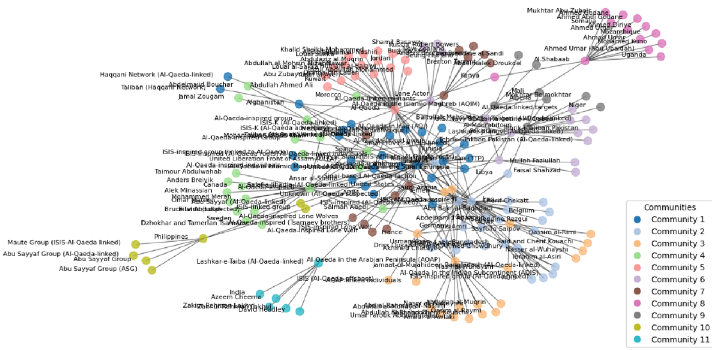

Fig. 3. Community Detection

early in the period, its influence diminished over time, with increased prominence of groups like **AQAP** and **ISIS-linked factions**. This temporal fluctuation reflects the network's adaptability in response to geopolitical changes and counterterrorism interventions, including leadership decapitations and regional realignments.

Concurrently, geospatial mapping in Fig. 4 using choropleth visualizations identified operational hotspots in **Pakistan**, **Somalia**, and the **United Kingdom**. These countries reported the highest incident frequencies and acted as central nodes in the country projection network, underscoring their strategic importance within Al-Qaeda's global operations. The combination of temporal and spatial insights highlights how terrorist networks evolve both structurally and geographically, informing future counterterrorism efforts.

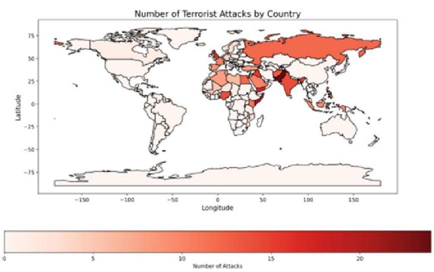

Fig. 4. Geospatial Analysis

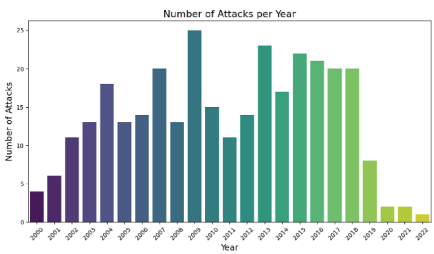

Fig. 5. Temporal analysis

4.4 Bipartite and Link Prediction Insights

The bipartite graph revealed operational overlaps between perpetrator groups and countries. Its projection into single-mode networks indicated clusters of similarly targeted countries and groups with shared operational domains (Fig. 6).

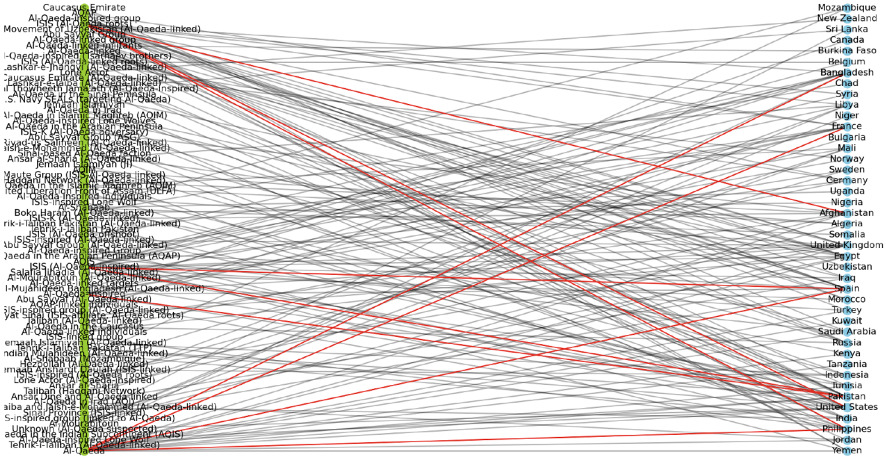

Fig. 6. Bipartite Graph with top 10 predicted links (RED)

4.5 Shortest Path Analysis

The **perpetrator group projection** (Fig. 7) highlights emerging alliances among terrorist groups and identifies clusters that could indicate coordinated actions or shared resources. The **shortest path** (Fig. 8) between Pakistan and Afghanistan indicates a strategic conduit for terrorist operations. This path bridges major terrorist networks and can be a potential choke point for counterterrorism.

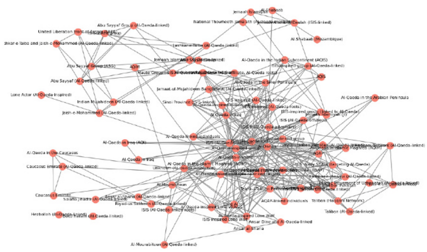

Fig. 7. Perpetrator Group Projection

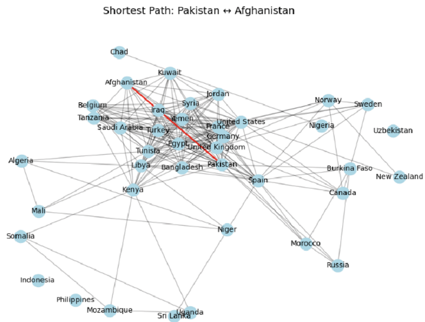

Fig. 8. Shortest Path Between Pakistan and Afghanistan

5 Conclusion and Future Scope

The limitations of this study stem primarily from the reliance on open-source datasets, which may contain gaps, inconsistencies, or underreporting, particularly in regions with limited media freedom. Additionally, the analysis was constrained by limited **geospatial resolution**—restricted to country-level data—and employed only a single link prediction algorithm (Adamic/Adar), which may not fully capture the complexity of evolving terrorist networks. Furthermore, real-time dynamics and qualitative factors such as ideological shifts, leadership motivations, or recruitment patterns were not fully explored.

The future scope of this research includes expanding the methodological framework by integrating multiple link prediction models, such as Jaccard coefficient, preferential attachment or machine learning approaches, to enhance predictive accuracy. Incorporating more granular geospatial data at the city or district level would allow for improved hotspot detection. Real-time data ingestion and dynamic network modeling could offer live monitoring and response capability. Applying this comprehensive SNA framework to other terrorist organizations would enable comparative analysis and contribute to the broader field of counterterrorism studies.

In conclusion, this study has presented a multi-layered Social Network Analysis (SNA) of Al-Qaeda and its affiliates from 2000 to 2020, using structural, temporal, bipartite, link prediction, and geospatial analyses to uncover the group's networked structure, shifting influence, and regional concentrations. By identifying central nodes, operational overlaps, and likely future links, the research enhances both theoretical understanding and provides practical tools for policymakers and intelligence agencies to anticipate and disrupt emerging threats.

References

1. Krebs, V.E.: Mapping networks of terrorist cells. Connections **24**(3), 43–52 (2002)
2. Gunaratna, R.: Inside Al Qaeda: Global Network of Terror. Columbia University Press (2002)
3. Hoffman, B.: Inside Terrorism. Columbia University Press (2006)
4. Everton, S.F.: Network analysis of terrorist organizations. J. Strateg. Stud. **35**(5), 597–622 (2012)
5. Xu, J., Chen, H.: Criminal network analysis and visualization. Commun. ACM **48**(6), 100–107 (2005)
6. LaFree, G., Dugan, L.: Introducing the global terrorism database (GTD). Terror. Polit. Violence **21**(1), 105–115 (2009)
7. RAND Corporation. RAND Database of Worldwide Terrorism Incidents (2020). https://www.rand.org/nsrd/projects/terrorism-incidents.html
8. National Consortium for the Study of Terrorism and Responses to Terrorism (START). Global Terrorism Database (GTD) (2020). https://www.start.umd.edu/gtd/

9. National Counterterrorism Center. Annual Reports and Country Reports on Terrorism (2020). https://www.dni.gov/nctc/index.html
10. United Nations Security Council. Reports on Global Terrorism and Sanctions Lists (2020). https://www.un.org/securitycouncil/ctc/reports